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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/977,072	05/11/2018	Chandra Jalluri	83950618	5563

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FGTL
Burris Law, PLLC
300 River Place Drive, Suite 1775
Detroit, MI 48207

EXAMINER	
TRAN, VINCENT HUY	

ART UNIT	PAPER NUMBER
2116	

NOTIFICATION DATE	DELIVERY MODE
12/02/2019	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

docketing@burrisiplaw.com

Office Action Summary	Application No. 15/977,072	Applicant(s) Jalluri et al.	
	Examiner VINCENT H TRAN	Art Unit 2116	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 5/11/2018.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,5-7,9-13 and 15-18 is/are rejected.

8) ☑ Claim(s) 2-4,8,14 and 19-20 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 5/11/2018 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 5/11/2018.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 103

2. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

3. The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

4. Claim 1, 9, 13, 16 is/are rejected under 35 U.S.C. 103 as being unpatentable over Yedatore et al. US Pub. No. 2011/0137697 ("Yedatore") in view of Chauvet et al. US Pub. No. 2018/0316729 ("Chauvet").

5. As per claim 1, Yedatore teaches a method for monitoring health of a machining tool [**see fig. 1**] operable in a machining state to form a workpiece and a non-machining state, the method comprising:

performing, by a control system, a machine health routine [**par. 0022**];

retrieving, by the control system, a non-machining cyclic schedule, in response to a corrective/failure event from the machine health routine, wherein the non-machining cyclic schedule defines a timetable for performing one or more maintenance routines [**see fig. 3A**];

identifying, by the control system, a primary maintenance routine from among the one or more maintenance routines based on the non-machining cyclic schedule; and

> [0022] **Tool 110 can send current performance characteristics of tool 110 to diagnostic server 120** and security server 140. Supplier server 180 can send a model of tool 110 to diagnostic server 120 through security server 140. A model can specify performance characteristics of tool 110 that correspond to normal behavior of tool 110. Diagnostic server 120 can create notifications for tool 110 in accordance with the model of tool 110. In one embodiment, the models of tool 110 may be provided by a field engineer or may be created based on knowledge of the field engineer. In an alternate embodiment, the models of tool 110 may be created by performing a simulation of a failure in a lab. In another alternate embodiment, the models of tool 110 may be created by a supplier of the tool and provided to diagnostic server 120 by supplier server 180 via security server 140. In one embodiment, diagnostic server monitors performance of tool 110 and compares current performance characteristics of tool 110 with the model of tool 110. **Upon detecting a deviation of any performance characteristic from the model 110, diagnostic server 120 creates a notification. The notification can specify one or more failures that are predicted to occur for tool 110 and may include a date or a number of days by which the failure is expected to occur**. Once notifications for tool 110 are created, diagnostic server 120 passes the notifications to automated corrective and predictive maintenance system 130.
>
> [0023] **Upon receiving a notification predicting a failure of tool 110 from diagnostic server 120, automated corrected and predictive maintenance system 130 can obtain maintenance data including one or more maintenance events for tool 110** from persistent storage (not shown) accessible to automated corrected and predictive maintenance system 130. **The maintenance data can also include a date or a number of days until the maintenance must be performed for each of the maintenance**

events. Maintenance events may include, for example**, a date by which the tool must be cleaned, a date by which the fluid levels within the tool must be topped off, a date by which a part in the tool must be replaced, a date by which firmware or software for the tool must be updated**, etc.

[0024] **Automated corrective and predictive maintenance system 130 can merge the failure event(s) indicated in the notification and the maintenance events to create an optimal schedule for the tool with an optimal set of options to schedule the known maintenance events for tool 110 and the failure events for tool 110 that occur within a predetermined amount of time from each other in order to minimize the down time of tool 110**. The optimal schedule can be used by automated corrective and predictive maintenance system 130 to notify support team system 160 of needed scheduled and unscheduled maintenance of tool 110. The optimal schedule can be used by predictive maintenance system 130 to notify vendor system 170 for parts required to perform maintenance on tool 110. In one embodiment, prior to notifying vendor 170, the required parts information is stored in security server 140.

[0045] FIG. **3A illustrates exemplary maintenance data used by an automated corrective and predictive maintenance system**. Maintenance data 300 includes maintenance events for a tool (e.g., 305, 310, 315, and 320) and a date associated with each maintenance event (e.g., 325, 330, 335, and 340). Clean tool 305 must be performed by date "01/01/2010" 325. Replace part 310 must be performed by date "12/31/2010" 330. Update software 315 must be performed by date "01/15/2010" 335. Update firmware 320 must be performed by date "01/30/2010" 340.

[0046] FIG. **3B illustrates exemplary notification data used by an automated corrective and predictive maintenance system**. Notification data 350 includes failure events indicating a prediction of a failure for a tool (e.g., 355 and 360) and a date associated with each failure event (e.g., 365 and 370). **Clean chamber X 355 must be performed by date "01/01/2010"** 365. Part Y fail 360 will occur by date "01/17/2010" 370, and therefore preventative action should be taken prior to that date to avoid the failure of part Y.

[**see further paragraph 0048-0051 and fig. 6**]

scheduling and performing, by the control system, the corrective/failure event with the primary maintenance routine [**see fig. 4**].

Yedatore does not teach determining, by a control system, whether a machine health routine is to be performed based on a machine health record stored by the control system; and

scheduling and performing, by the control system, the machine health routine with the primary maintenance routine.

Chauvet teaches determining, by a control system, whether a machine health routine is to be performed based on a machine health record stored by the control system; and scheduling and performing, by the control system, the machine health routine with the primary maintenance routine.

> [0202] Referring to FIGS. 17 and 18, ***a diagnostic event 1818C***, in some embodiments, can be associated with any components of the SDA system including, for example, compute nodes, network components (e.g., switches), and the like. ***A diagnostic event is typically triggered when a predefined condition is met***. For example, ***when an equipment has reached its time limit for continuous operation, when an equipment does not reach a certain position on time, or when network delay exceeds a certain time***. In some embodiments, a diagnostic event can be triggered by an external signal. For example, an analytics engine running on a cloud (e.g., cloud 450 in FIG. 4) ***that can gather data including monitoring data from the field and convert into actionable information such as diagnostic information real time. Such an engine can generate a signal when diagnostic data indicates potential problem. A diagnostic event detection component (e.g., event detection component 1726 in FIG. 17) can detect the diagnostic event and in response, a diagnostic event handler (e.g., component 1820C in FIG. 18B) can either schedule or perform a diagnostic check of the component that triggered the diagnostic event***. In some embodiments, the diagnostic event handler can coordinate with the orchestration components to facilitate the diagnostic check of the component. For example, if a network switch has a diagnostic event, then the diagnostic event handler can request the network controller (e.g., via the network orchestration component 1922 in FIG. 19 and/or system orchestration component 1916 in FIG. 19) to redirect network flows away from that network switch while diagnostic checks are performed on it. ***In some embodiments, a diagnostic event can trigger another event such as a maintenance event or an upgrade event.***
>
> [0203] Another type of event that the event detection component 1726 can detect is a ***maintenance event 1818D. A maintenance event can be scheduled in advance, initiated on demand by a user to inspect and/or repair one or more compute nodes or in response to other events such as diagnostic events***. At the scheduled time or in response to a user request, a maintenance event can be triggered and detected by the event detection component. In response to the event, a maintenance event handler 1820 can be invoked. The maintenance event handler can utilize the fog server orchestration component to shift application processes from a compute node scheduled to undergo maintenance to another compute node (e.g., virtual machines, containers or bare metals). The maintenance event handler can also, via the network orchestration component and the CS orchestration component, remediate or reconfigure the network and the security

> environments to enable the virtualized application functions to control a machine or process. In some embodiments, one example response to a maintenance event can be similar to a response to a compute node fault event described in reference to FIG. 20.

At the time of the invention was filed, it would have been obvious to one of ordinary skill in the art to combine the teachings of the cited references since they both directed to the system and method for monitoring health of a system. Chauvet teaches determining whether a machine health routine is to be performed based on a machine health record stored by the control system; and scheduling and performing, by the control system, the machine health routine with the primary maintenance routine would further improve the efficiency of the system by provide the system the ability to schedule health, corrective, and maintenance routine during the same period to reduce overall down time without require the system to continuously perform diagnostic routine even when the system is healthy.

6. As per claim 9, Yedatore in view of Chauvet teaches placing, by the control system, the machining tool in the non-machining state from the machining state when the primary maintenance routine is to be performed [see par. 0017 of Yedatore]; performing, by the control system, the primary maintenance routine and the machine health routine [see discussion in claim 1]. Yedatore in view of Chauvet does not teach storing, by the control system, data indicative of a date and time that the maintenance health routine was performed as part of the machine health record. However, examiner takes official notice such feature of saving date and time that the maintenance health routine was performed was old and well known in the art of maintenance process. One of ordinary skill in the art would motivated to provide such feature in order to allow the system the ability to accurately determine the next schedule.

7. As per claim 13, Yedatore teaches a machining tool comprising:

a machine operable in a machining state to machine a workpiece and in a non-machining state to perform a maintenance routine from among a plurality of maintenance routines; and

a control system configured to control the machine in the machining state and the non-machining state, wherein the control system is configured to execute a corrective routine in conjunction with a primary maintenance routine selected from among the plurality of maintenance routines based on a non-machining cyclic schedule that defines dates and times for performing the maintenance routines

and Chauvet [**par. 0202 and 0203**] teaches control system is configured to execute a machine health routine in conjunction with a primary maintenance routine and a machine health record that includes information indicative of a subsequent test time for performing the machine health routine [**see further discussion in claim 1**].

8. As per claim 16, Yedatore/Chauvet teaches primary maintenance is a maintenance routine that is to be performed at a specific timeslot defined by the non-machining cyclic schedule and is proximal to the subsequent test time of the machine health routine [see discussed par. in claim 1].

9. Claim 17 is/are rejected under 35 U.S.C. 103 as being unpatentable over Yedatore in view of Chauvet and further in view of Dister et al. U.S. Patent No. 6,289,735 ("Dister").

10. As per claim 17, Yedatore in view of Chauvet teaches a machining tool control method comprising:

determining a performance time for executing an analysis on a machining tool based on a test frequency for the analysis [*Chauvet in par. 0202 - A diagnostic event is typically triggered*

when a predefined condition is met… when an equipment has reached its time limit for continuous operation…];

retrieving a non-machining schedule that defines a timetable for executing multiple maintenance routines on the machining tool [*see fig. 3A of Yedatore*];

selecting, from among the maintenance routines, a primary routine that is performed at a time period proximal to the performance time; and

executing the analysis with the primary routine [*see discussion in claim 1*].

Yedatore/Chauvet does not teach the analysis is a vibration analysis. However, such features of vibration analysis is old and well known in the art to machine health evaluation.

Dister teaches another diagnostic system for diagnosing the health of a machine. Specifically, Dister teaches the system includes vibration sensor mounted on a machine to measure vibration where the vibration signals from the sensor are processed and analyzed by the system.

> (6) **Vibration analysis is an established nonintrusive technique for measuring the health of mechanical components in rotating machines**. Every rotating machine exhibits a characteristic vibration signature which varies with the design, manufacture, application and wear of each component.
>
> (17) The system includes at least one vibration sensor mounted on the machine to sense vibrations. **The system evaluates vibration signals from the vibration sensor by employing a diagnostic module which processes and analyzes the vibration signals**.
>
> (34) Returning to FIG. 1a, once the processor 90 has processed all of the respective motor data, the processed data may be sent to the host computer 66 for analysis. **The host computer 66 may then make determinations as to the health of the motor 30 based on the data received from the diagnostic module 44. Accordingly, motor maintenance can be scheduled to correspond with the state of the motor 30**. Additionally, the processed data can be compiled and stored for trend analysis and forecasting. Since the diagnostic module 44 is integrated with the motor 30, the data sampling rate can be substantially high thus providing accurate and up-to-date data relating to the health of the motor 30. However, as mentioned above, motor diagnosis, trend analysis, forecasting, etc. that could be performed by the host computer 66 alternatively may be performed directly by the diagnostic module 44.

At the time of the invention was filed, it would have been obvious to one of ordinary skill in the art to modify the method of Yedatore/Chauvet with the vibration analysis of Dister. The motivation for doing so would has been to provide instant indication of the machinery condition where vibration analysis can quickly detect misalignment and unbalance in a machine before conditions result in bearing or shaft deterioration without overly intrusive.

11. As per claim

12. Claims 5-7 is/are rejected under 35 U.S.C. 103 as being unpatentable over Yedatore/Chauvet as applied to claim 1 above, and further in view of Shi US Pub. No. 2017/0236075.

13. As per claim 5, Yedatore/Chauvet does not teach the machine health record defines parameters for determining performance of the machine health routine and includes data indicative of an operation history of the maintenance health routine.

Shi teaches a system and method relates generally to machine inspection and maintenance, in particular, to an improved system and method for using data analysis to optimize machine inspection and maintenance. Specifically, Shi teaches the machine health record defines parameters for determining performance of the machine health routine and includes data indicative of an operation history of the maintenance health routine [par. 0009, 0031].

At the time of the system was filed, it would have been obvious to one of ordinary skill in the art to modify the system of Yedatore/Chauvet with the machine health record defines parameters for determining performance of the machine health routine and includes data indicative of an operation history of the maintenance health routine. the motivation for doing so

would has been to provide the machine the ability to assess the operate condition and health of the machine in timely fashion. Thus improve accuracy with diagnosis of problems and timeliness of the appropriate responses to such issues.

14. As per claim 6, Chauvet teaches determining and storing, by the control system, a subsequent test time for performing the maintenance health routine based on the parameters and the operation history defined by the machine health record, wherein: the determining whether the machine health routine is to be performed further comprises: comparing the subsequent test time provided in the machine health record with an internal clock of the control system, and determining the machine health record is to be performed in response to the subsequent test time being within a predefined time period [see par. 0202].

15. As per claim 7, Yedatore in view of Chauvet teaches the primary maintenance routine is performed at a specific timeslot defined by the non-machining cyclic schedule and is proximal to the subsequent test time of the machine health routine [see par. discussed in claim 1].

16. Claims 10-12, 15 is/are rejected under 35 U.S.C. 103 as being unpatentable over Yedatore/Chauvet as applied to claim 1 or 13 above, and further in view of Al-Gouhi et al. US Pub. No. 2017/0356283 ("Al-Gouhi").

17. As per claim 10, Yedatore/Chauvet does not teach setting a test frequency for the machine health routine to a first frequency for a first operation period; setting the test frequency for the machine health routine to a second frequency having a greater duration than that of the first frequency in response to the lapse of the first operation period; and setting the test frequency for the machine health routine to the first frequency in response to the machining tool experiencing an abnormal operation.

Al-Gouhi teaches a method of automated preventive and predictive maintenance for a system. Specifically, Al-Gouhi teaches setting a test frequency for the machine health routine to a first frequency for a first operation period; setting the test frequency for the machine health routine to a second frequency having a greater duration than that of the first frequency in response to the lapse of the first operation period; and setting the test frequency for the machine health routine to the first frequency in response to the machining tool experiencing an abnormal operation [par. 0039 *-diagnostic data may need to be captured more frequently to avoid a valve failure if the valve is beginning to wear. For example, if normal maintenance occurs every three months, the PMM 114 can determine the reliability of Probability(x>3 months). If this reliability is high, say, 90%, the 3-month normal maintenance cycle for capturing diagnostic data can be sufficient for the valve. In some cases, if the valve is in good condition, a maintenance cycle duration longer than the normal cycle duration can be used to reduce maintenance cost. However, if the reliability is low, say, 10%, an increased frequency of capturing diagnostic data, for example, once every month, may be needed. The PMM 114 can send the reliability to the WFM 110 so that the WFM 110 can dynamically schedule maintenance for each valve based on individual health condition*.]

At the time of the invention was filed, it would have been obvious to one of ordinary skill in the art to modify the method of Yedatore/Chauvet with the feature discussed above of Al-Gouhi. The motivation for doing so would have been to reduce system downtime due to frequent health check.

18. As per claim 11, Al-Gouhi teaches the first operation period is based on a cycle run life of a component of the machining tool [par. 0039].

19. As per claim 12, Al-Gouhi teaches the first operation period is based on a cycle run life of a component of the machining tool [par. 0039].

20. As per claim 15, Al-Gouhi teaches he control system is configured to set a test frequency for performing the machine health routine, and the control system sets the test frequency to a first frequency for a first operation period, to a second frequency longer than that of the first frequency in response to the lapse of the first operation period; and to the first frequency in response to an abnormal operation of the machine [par. 0039].

21. Claim 18 is/are rejected under 35 U.S.C. 103 as being unpatentable over Yedatore/Chauvet/Dister as applied to claim 17 above, and further in view of Al-Gouhi.

22. As per claim 18, Al-Gouhi teaches setting the test frequency to a second frequency having a greater duration than that of the first frequency in response to the lapse of the first operation period; and setting the test frequency to the first frequency in response to the machining tool experiencing an abnormal operation [see discussion in claim 10].

Allowable Subject Matter

Claims 2-4, 8, 14, 19-20 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to VINCENT HUY TRAN whose telephone number is (571)272-7210. The examiner can normally be reached on M-F 7:00-4:00.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Kenneth M Lo can be reached on 5712729774. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/VINCENT H TRAN/
Primary Examiner, Art Unit 2121